

Inverse design of membrane-type acoustic metamaterial based on deep learning

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ABSTRACT

Membrane acoustic metamaterial (MAM) design mainly relies on forward design methods, making it difficult to determine structural parameters directly from predefined acoustic requirements. In this study, an inverse design framework for rubber-metal composite mass-block MAM (RMMAM) is proposed. A LiveLink-based COMSOL–MATLAB simulation interface is established to automatically generate finite element datasets, and a dual attention convolutional neural network based on the EfficientNet-feature pyramid architecture is trained to predict structural parameters from target sound transmission loss (STL) curves. Unlike conventional forward strategies, the proposed framework directly maps target STL curves to the corresponding RMMAM parameters. With target STL curves as input, the prediction error of structural parameters is controlled within 10.5%. In addition, STL curves reconstructed from the predicted parameters exhibit peak-frequency errors of 5–20 Hz and STL errors of 1.13–10.92 dB, demonstrating the reliability of the inverse design model. Case studies are further conducted under the constraint that the peak sound insulation level should not be lower than 50 dB in the 300–400 Hz frequency band. Both numerical simulations and experiments confirm that the designed RMMAM structures satisfy the predefined sound insulation requirements. The proposed framework provides an efficient route for the inverse design of acoustic metamaterials and supports their engineering applications.

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I. INTRODUCTION

Acoustic metamaterial (AM) have attracted considerable attention in various scientific and engineering fields owing to their exceptional acoustic properties.^{1,2} In recent years, significant progress has been achieved in the theoretical modeling and practical applications of AM across multiple disciplines, including physics and mathematics.^{3–6} Membrane acoustic metamaterial (MAM), which consist of a rigid frame, a mass block, and a thin membrane,^{7,8} were first proposed by Yang *et al.*⁹ Owing to their unique structural configuration and localized resonance mechanisms, MAM exhibit excellent low-frequency sound insulation performance and have been widely applied in underwater acoustics,¹⁰ aerospace engineering,¹¹ and other related fields.

Numerical optimization-based inverse design methods have also been widely applied to the structural design of acoustic metamaterials. Topology optimization can directly search for material distributions or structural topologies that satisfy target acoustic performance under prescribed design domains, material constraints, and objective functions. For example, Noguchi

*et al.*¹² designed hyperbolic acoustic metamaterials using a high-frequency homogenization method combined with level-set topology optimization, enabling the control of anisotropic acoustic propagation. Fujii *et al.*¹³ applied topology optimization to acoustic–elastic coupled systems and designed an acoustic cloak. They further proposed a topology-optimized source-shifter for camouflaging the location of a sound source.¹⁴ In addition, Fujii *et al.*¹⁵ designed labyrinthine acoustic metamaterials based on level-set topology optimization to achieve broadband negative refractive-index characteristics. Subsequently, they proposed a topology-optimization-inspired labyrinthine acoustic metamaterial with a subwavelength bandgap.¹⁶ These studies demonstrate that conventional numerical optimization methods, such as topology optimization, can generate complex and even non-intuitive acoustic metamaterial structures with a high degree of design freedom.

However, although topology optimization methods have strong structural search capabilities, they usually require repeated finite element calculations and iterative optimization. When the target acoustic response or design conditions change, the optimization

process often needs to be restarted, resulting in high computational cost and limited efficiency for large-scale dataset generation and rapid inverse design. With the development of computer science, machine learning and deep learning have gradually become effective tools for the inverse design of acoustic metamaterials.^{17–19} By leveraging machine learning and deep learning, structural parameters and geometrical configurations that meet the desired acoustic performance can be inversely inferred from predefined target responses.²⁰ To establish the mapping relationship among structural parameters, geometrical configurations, and acoustic performance, the construction of datasets is of critical importance. For instance, Gao *et al.*²¹ employed sound absorption coefficients generated using Latin hypercube sampling as training data for machine learning models. Huang *et al.*²² utilized parametric models and calculated bandgap characteristics through finite element simulations to construct datasets for deep learning. Similarly, Liu *et al.*²³ adopted sound absorption coefficient curves obtained from finite element simulations under different structural parameters as training samples. Although these approaches ensure the accuracy and reliability of dataset construction, relying solely on finite element simulations to generate large-scale datasets is computationally expensive and time-consuming, which significantly limits the efficiency of inverse design.

Mechanical learning and deep learning have derived many kinds of network types,^{24–26} and as AM and neural networks gradually become mature, more and more scholars have referenced neural networks to AM. Gurbuz *et al.*²⁷ proposed a generative adversarial network (GAN) single-cell generation framework for AM, which fuses structural geometries and transmission loss features through multiple input channels to realize the ideal transmission characteristics. Sun *et al.*²⁸ combined the lumped parameter method and deep neural network (DNN) to realize an inverse design method for structures such as multistep resonators by introducing equivalent electrical parameters to establish a link between structure and performance. Liu *et al.*²⁹ constructed a convolutional neural network (CNN) network to realize the inverse design method for structures in the 860–8000 Hz frequency range with an average absorption in the range of 97%.

To investigate the inverse design methodology for rubber-metal composite mass-block MAM (RMMAM), as well as efficient dataset generation and feature extraction for neural network models, an efficient feature pyramid network with dual attention (EFDA)-CNN framework was developed in this study. Three types of RMMAM structures with different composite mass-block distributions were constructed, and the ranges of geometric and material parameters were determined. Through the LiveLink interface, a coupled COMSOL and MATLAB simulation platform was established to generate the STL dataset automatically. Using the resulting STL curves as input data, EfficientNet-feature pyramid network (FPN) was adopted as the backbone network, and a dual attention mechanism was incorporated to enhance image-feature extraction, thereby enabling the mapping from STL characteristics to the corresponding structural parameters.

II. SOUND INSULATION PERFORMANCE OF RMMAM STRUCTURES AND DATASET ESTABLISHMENT

A. RMMAM structure modeling

The RMMAM is composed of a square frame structure, a tensioned membrane, and three composite mass blocks. In the

construction of the model, the square frame provides well-defined boundary conditions and stress distribution. The membrane is tensioned within the square frame to form a vibrating medium. Three two-layer composite mass blocks are attached to the membrane surface, forming a multi-degree-of-freedom vibration system. Each composite mass block consists of an upper metal mass block and a lower rubber mass block. To improve the ability of the inverse design network to characterize the sound insulation properties under different geometric configurations and to enhance the adaptability of the model to different structural configurations, three different mass-block distributions are proposed. As shown in Fig. 1, Fig. 1(a) illustrates the three mass-block distribution structures. The mass blocks in structure 1 are diagonally arranged. The mass blocks in structure 2 are horizontally arranged. The mass blocks in structure 3 are arranged in a T-shaped distribution. Figure 1(b) shows a schematic diagram of the structure with a single composite mass block. The STL curves generated from these three structural configurations are used to construct a dataset to establish the mapping relationship between structural parameters, material parameters, and STL performance.

B. RMMAM theoretical modeling of STL

When a planar acoustic wave is incident normally on the surface of the RMMAM structure, part of the acoustic energy is reflected, part is absorbed, and the remaining energy propagates through the structure. The acoustic transmission characteristics of the structure can be characterized by the acoustic pressure transmission coefficient, which is defined as the ratio of the transmitted acoustic power to the incident acoustic power. STL is further calculated from the acoustic pressure transmission coefficient t_p .

Based on the acoustic field impedance method and the transfer matrix method, the sound insulation performance of RMMAM structures with three different distributions is investigated. Without considering the localized mass blocks, the equation of motion of the membrane under external pressure is given by

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{H}p_{\text{diff}} \quad (1)$$

where $\mathbf{q}$ is the modal generalized coordinates of the membrane, $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the modal mass, damping, and stiffness matrices of the membrane, respectively, $\mathbf{H}$ is the coupling vector, and $p_{\text{diff}} = p^-(0) - p^+(0)$ is the pressure difference between the two sides.

In the frequency domain,

$$(\mathbf{K} + j\omega\mathbf{C} - \omega^2\mathbf{M})\mathbf{q} = \mathbf{H}p_{\text{diff}}. \quad (2)$$

Define the dynamic stiffness matrix,

$$\mathbf{D}_{\text{mem}}(\omega) = \mathbf{K} + j\omega\mathbf{C} - \omega^2\mathbf{M}. \quad (3)$$

For the i th composite mass block (lower rubber block and upper metal block), two local degrees of freedom are taken. In

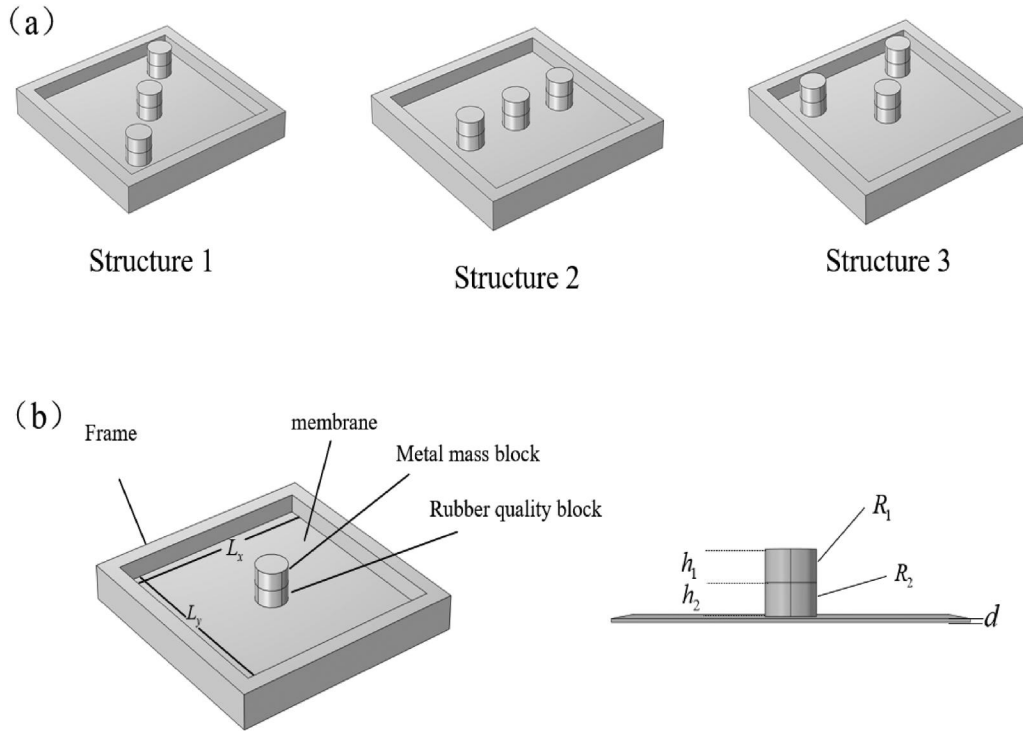


FIG. 1. RMMAM model. (a) The RMMAM model with different mass-block distributions. (b) Schematic of a single composite mass-block structure.

frequency, the equation can be written as

$$\mathbf{D}_{mass}^{(i)}(\omega) \begin{bmatrix} x_1^{(i)}(\omega) \\ x_2^{(i)}(\omega) \end{bmatrix} = \begin{bmatrix} f^{(i)}(\omega) \\ 0 \end{bmatrix}, \quad i = 1, 2, 3, \quad (4)$$

where $\mathbf{D}_{mass}^{(i)}(\omega)$ is the local mass-block stiffness matrix, $x_1^{(i)}$ is the normal displacement of the rubber block, $x_2^{(i)}$ is the normal displacement of the metal block, and $f^{(i)}(\omega)$ is the equivalent general force acting on the rubber block by the membrane.

Each composite mass is internally connected to the upper and lower composite masses by springs and damping, and the local dynamic stiffness matrix of the i th composite mass can be written as

$$\mathbf{D}_{mass}^{(i)}(\omega) = \begin{bmatrix} a_{11}^{(i)}(\omega) & a_{12}^{(i)}(\omega) \\ a_{21}^{(i)}(\omega) & a_{22}^{(i)}(\omega) \end{bmatrix} = \begin{bmatrix} k_1^{(i)} - \omega^2 m_1^{(i)} + j\omega c_1^{(i)} - k_c^{(i)} & -k_c^{(i)} \\ -k_c^{(i)} & k_2^{(i)} - \omega^2 m_2^{(i)} + j\omega c_2^{(i)} \end{bmatrix}, \quad (5)$$

where $m_1^{(i)}$ is the mass of the metal layer of the first composite block, $m_2^{(i)}$ is the mass of the rubber layer of the first composite block, and $k_1^{(i)}$, $c_1^{(i)}$ is the equivalent stiffness and damping between

the first rubber block and the membrane, since the upper metal block is not in direct contact with the membrane, and there are $k_2^{(i)}$ and $c_2^{(i)}$ zeros.

Since only the lower rubber block is directly attached to the membrane, $x_1^{(i)}$ is taken as the membrane-coupled local degree of freedom, whereas $x_2^{(i)}$ represents the internal degree of freedom of the upper metal block. Accordingly, the local dynamic equation in Eq. (4) can be written in component form as

$$\begin{cases} a_{11}^{(i)}(\omega)x_1^{(i)}(\omega) + a_{12}^{(i)}(\omega)x_2^{(i)}(\omega) = f^{(i)}(\omega), \\ a_{21}^{(i)}(\omega)x_1^{(i)}(\omega) + a_{22}^{(i)}(\omega)x_2^{(i)}(\omega) = 0. \end{cases} \quad (6)$$

By eliminating the internal degree of freedom $x_2^{(i)}(\omega)$ through static condensation (Schur complement), the local two degree of freedom system is reduced to an equivalent scalar relation associated with the membrane-coupled displacement $x_1^{(i)}(\omega)$. The formula is as follows:

$$d_{11}^{(i)}(\omega)x_1^{(i)}(\omega) = f^{(i)}(\omega), \quad (7)$$

where $\tilde{d}_{11}^{(i)}(\omega)$ denotes the equivalent dynamic stiffness of the i th composite mass block referred to the membrane-coupled lower rubber block. This equivalent mass incorporates the dynamic effects of the upper metal mass block into the response of the

lower degree of the freedom system and can be expressed as

$$d_{11}^{(i)}(\omega) = a_{11}^{(i)}(\omega) - \frac{a_{12}^{(i)}(\omega)a_{21}^{(i)}(\omega)}{a_{22}^{(i)}(\omega)}. \quad (8)$$

Equation (7) shows that the equivalent dynamic stiffness consists of the direct dynamic stiffness term $a_{11}^{(i)}(\omega)$ and a correction term caused by the coupling between the lower rubber block and the upper metal block. Therefore, the contribution of the internal degree of freedom $x_2^{(i)}$ is retained in the reduced model through the coupling term $a_{12}^{(i)}(\omega)$, $a_{21}^{(i)}(\omega)$, and $a_{22}^{(i)}(\omega)$, while the final condensed equation remains expressed only in terms of the variable $x_1^{(i)}(\omega)$.

Since only the lower mass block of each composite mass block is directly coupled to the membrane, and the upper metal block is not directly coupled, the coupling submatrix $\mathbf{B}^{(i)}(\omega)$ corresponding to the i th composite mass block is an $N \times 1$ column vector. Then, the column vector of the coupling between the lower degrees of freedom of the i composite mass block and the N modes in the membrane is

$$\mathbf{B}^{(i)}(\omega) \in \mathbb{C}^{N \times 1}, \quad i = 1, 2, 3. \quad (9)$$

Then, the equivalent dynamic stiffness matrix of the membrane is obtained after eliminating the local degrees of freedom for the three composite mass blocks as

$$\mathbf{D}_{eff}(\omega) = \mathbf{D}_{mem}(\omega) - \sum_{i=1}^3 \mathbf{B}^{(i)}(\omega) \left[d_{11}^{(i)}(\omega) \right]^{-1} \mathbf{B}^{(i)T}(\omega). \quad (10)$$

The generalized coordinates of the membrane modes without added mass blocks can be obtained using Eqs. (2) and (3),

$$\mathbf{q} = \mathbf{D}_{mem}^{-1}(\omega) \mathbf{H} p_{diff}. \quad (11)$$

After introducing three composite mass blocks and eliminating their local degrees of freedom, the original dynamic stiffness matrix of the membrane is updated from $\mathbf{D}_{mem}(\omega)$ to $\mathbf{D}_{eff}(\omega)$, replacing $\mathbf{D}_{eff}(\omega)$ by $\mathbf{D}_{mem}(\omega)$, obtain the model generalized coordinates of the membrane for the three composite mass blocks,

$$\mathbf{q}_1 = \mathbf{D}_{eff}^{-1}(\omega) \mathbf{H} p_{diff}, \quad (12)$$

where $\mathbf{q}_1$ represents the generalized coordinates of the membrane state of the additional mass block.

The average normal velocity of the membrane is expressed modally as

$$v = j\omega \frac{1}{S} \mathbf{H}^T \mathbf{q}_1 = j\omega \frac{1}{S} \mathbf{H}^T \mathbf{D}_{eff}^{-1}(\omega) \mathbf{H} p_{diff}, \quad (13)$$

where S is the acoustic area of the membrane.

The surface mechanical impedance derived from the above equation is defined as

$$Z_{RMMAM}(\omega) = \frac{p_{diff}}{v} = \frac{S}{j\omega \mathbf{H}^T \mathbf{D}_{eff}^{-1}(\omega) \mathbf{H}}. \quad (14)$$

The overall transfer matrix can be expressed as

$$\mathbf{T} = \mathbf{T}_a \mathbf{T}_s, \quad (15)$$

where $\mathbf{T}_a$ denotes the transfer matrix of the air layer and $\mathbf{T}_s$ denotes the transfer matrix of the three composite mass-block structural layers.

Since the constitutive layer is an air layer, the transfer matrix $\mathbf{T}_a$ is expressed as

$$\mathbf{T}_a = \begin{bmatrix} \cos(k_z d_0) & j\rho_a c_a \sin(k_z d_0) \\ \frac{j \sin(k_z d_0)}{\rho_a c_a} & \cos(k_z d_0) \end{bmatrix}, \quad (16)$$

where k_z is the wave number component perpendicular to the surface of the constituent layer, $k_z = k_a \cos \alpha$, α is the azimuthal angle of the incident wave, d_0 is the cavity spacing, $Z_a = \rho_a c_a$, Z_a is the acoustic impedance in the air, ρ_a is the density of the air, and c_a is the speed of sound in the air.

The transfer matrix $\mathbf{T}_s$ can be expressed as

$$\mathbf{T}_s = \begin{bmatrix} 1 & Z_{RMMAM}(\omega) \\ 0 & 1 \end{bmatrix}. \quad (17)$$

The four elements T_{11} , T_{12} , T_{21} , and T_{22} in $\mathbf{T}$ can be obtained by Eq. (15), which are denoted as

$$\begin{cases} T_{11} = \cos(k_z d_0), \\ T_{12} = \cos(k_z d_0) Z_{RMMAM} + j\rho_a c_a \sin(k_z d_0), \\ T_{21} = \frac{j \sin(k_z d_0)}{\rho_a c_a}, \\ T_{22} = \frac{j Z_{RMMAM}}{\rho_a c_a} \sin(k_z d_0) + \cos(k_z d_0). \end{cases} \quad (18)$$

The acoustic pressure transmission coefficient t_p of the three composite mass-block structures can be obtained from the four elements of the transfer matrix and is expressed as

$$t_p = \left| \frac{2}{T_{11} + T_{12} \cos \alpha / (\rho_a c_a) + T_{21} (\rho_a c_a) / \cos \alpha + T_{22}} \right|. \quad (19)$$

Then, the equation for STL is expressed as

$$\text{STL} = 20 \log_{10} \left| \frac{1}{t_p} \right|. \quad (20)$$

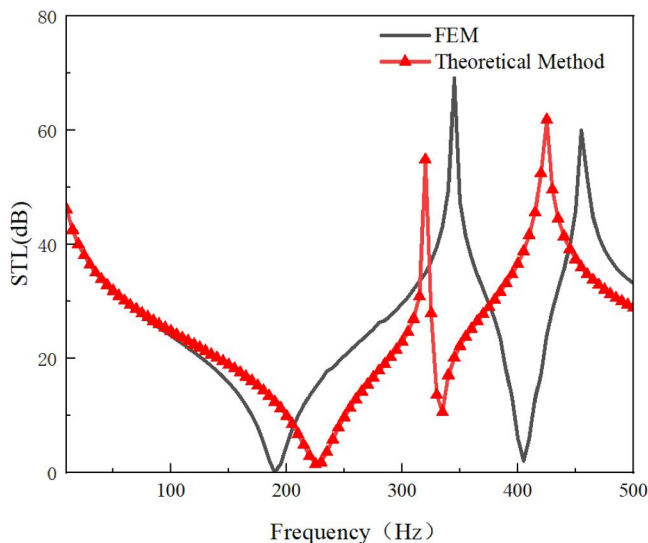
C. STL result analysis

To investigate the acoustic performance of the RMMAM structure, the theoretical and simulation results of STL curves are

TABLE I. Dimensional and material parameters of RMMAM.

RMMAM	Material and dimensional parameters	parameter value
Frameworks	Density ρ	1160 kg/m ³
	Young's modulus E	2×10^9 Pa
	Poisson's ratio ν	0.375
Top metal block	Height h_1	2 mm
	Radius R_1	2 mm
	Density ρ_1	8960 kg/m ³
	Young's modulus E_1	11×10^{10} Pa
	Poisson's ratio ν_1	0.34
Lower rubber block	Height h_2	2 mm
	Radius R_2	2 mm
	Density ρ_2	2000 kg/m ³
	Young's modulus E_2	1.8873×10^6 Pa
	Poisson's ratio ν_2	0.5
Membrane	Thicknesses d	0.2 mm
	Density ρ_3	1500 kg/m ³
	Young's modulus E_3	0.36×10^9 Pa
	Poisson's ratio ν_3	0.32
	Length L_x and L_y	26 mm

calculated. The parameters are set as shown in Table I. The mass-block distribution is diagonal, the air density $\rho_0 = 1.21$ kg/m³, and the speed of sound in air $c_0 = 343$ m/s. As can be seen in Fig. 2, the first peak of the finite element method is at 345 Hz, and the second peak is at 455 Hz. The first peak of the theoretical method is at 320 Hz, and the second peak is at 425 Hz, and the detailed data are shown in Table II. Table II shows that the STL curves

**FIG. 2.** STL of the RMMAM structure.**TABLE II.** Simulation and analytical STL results of the RMMAM structure.

Peak location	Peak sound insulation frequency (Hz)			Peak sound insulation (dB)		
	FEM	Theoretical	Error	FEM	Theoretical	Error
First peak	345	320	-25	69.15	54.81	-14.34
Second peak	455	425	-30	60.06	61.87	-1.81

obtained by the two methods exhibit similar trends, demonstrating good agreement between the theoretical predictions and the finite element results. However, a slight discrepancy is observed at the resonance peak frequency. This difference arises because, in the theoretical model, the elimination of local degrees of freedom introduces an equivalent damping, which represents an idealized damping state. In contrast, the damping considered in the finite element method is non-ideal, resulting in a small deviation in the peak frequency, although both methods remain consistent in terms of the overall trend and the locations of peaks and valleys.

D. Dataset establishment and parameter selection range

To obtain an accurate dataset, this paper utilizes LiveLink for MATLAB to establish a simulation interface between COMSOL Multiphysics 6.2 and MATLAB. Physical models created in COMSOL Multiphysics 6.2 are directly called upon within MATLAB for simulation; the model automatically performs geometric reconstruction, mesh updating, frequency analysis, and result export in response to changes in code parameters. The sampling process proceeds sequentially according to changes in the material parameters; each set of parameter changes triggers a simulation, which outputs the corresponding STL sound insulation curve, ultimately forming a complete dataset. Compared with the traditional method of setting parameters, running simulation and exporting results in COMSOL, which is carried out by humans, the time is greatly reduced. The material parameter ranges are shown in Table III. To ensure that the density of the metal mass blocks can be correctly obtained during the experiment, the densities of five different commonly used metal mass blocks are chosen in this paper, namely, copper, iron, aluminum, zinc, and magnesium.

III. RMMAM INVERSE DESIGN BASED ON EFDA-CNN

For the inverse design of RMMAM, the STL curves are used as the neural network dataset. The curves are converted

TABLE III. Parameter range of RMMAM.

Parameters	R_1 and R_2 (mm)	d (mm)	ρ_1 (kg/m ³)	ρ_2 (kg/m ³)	ρ_3 (kg/m ³)
Range	(1, 3)	(0.1, 0.4)	(1740/2700/7140/7870/8960)	(1000, 2000)	(1000, 2000)

into three-channel images to serve as the network's input. To ensure the accuracy of prediction, EfficientNet is used in the CNN and paired with an FPN as the backbone network. Subsequently, the extracted features are enhanced by the dual attention mechanism and transformed into one-dimensional feature vectors through global pooling, which are then fed into the fully connected layer. A weighted loss function is employed to train the network, and the best-trained model is finally used for prediction.

A. EFDA-CNN neural network framework

1. EfficientNet-FPN network

EfficientNet-FPN serves as the backbone network of the entire model, and its overall structure is shown in Fig. 3. The main body of EfficientNet-FPN consists of five stages of inverted

residual convolution (MBConv). The MBConv block primarily comprises convolution layers, depthwise separable convolutions, and Squeeze-and-Excitation (SE) attention blocks. The first standard convolution is followed by batch normalization (BN) and the Swish activation function. It is mainly used to expand the channel dimension of the input features, mapping low-dimensional features to a higher-dimensional space and enhancing the feature representation capability of the network. Subsequently, the depthwise separable convolution is also followed by BN and the Swish activation function. This design allows each channel to perform convolution independently, which effectively captures local spatial features while reducing the computational load of the network. Next, after the depth wise separable convolutional layer, the feature map is fed into the SE attention module. This module consists of two branches. The upper branch preserves the original features, while the lower branch applies global pooling to the feature

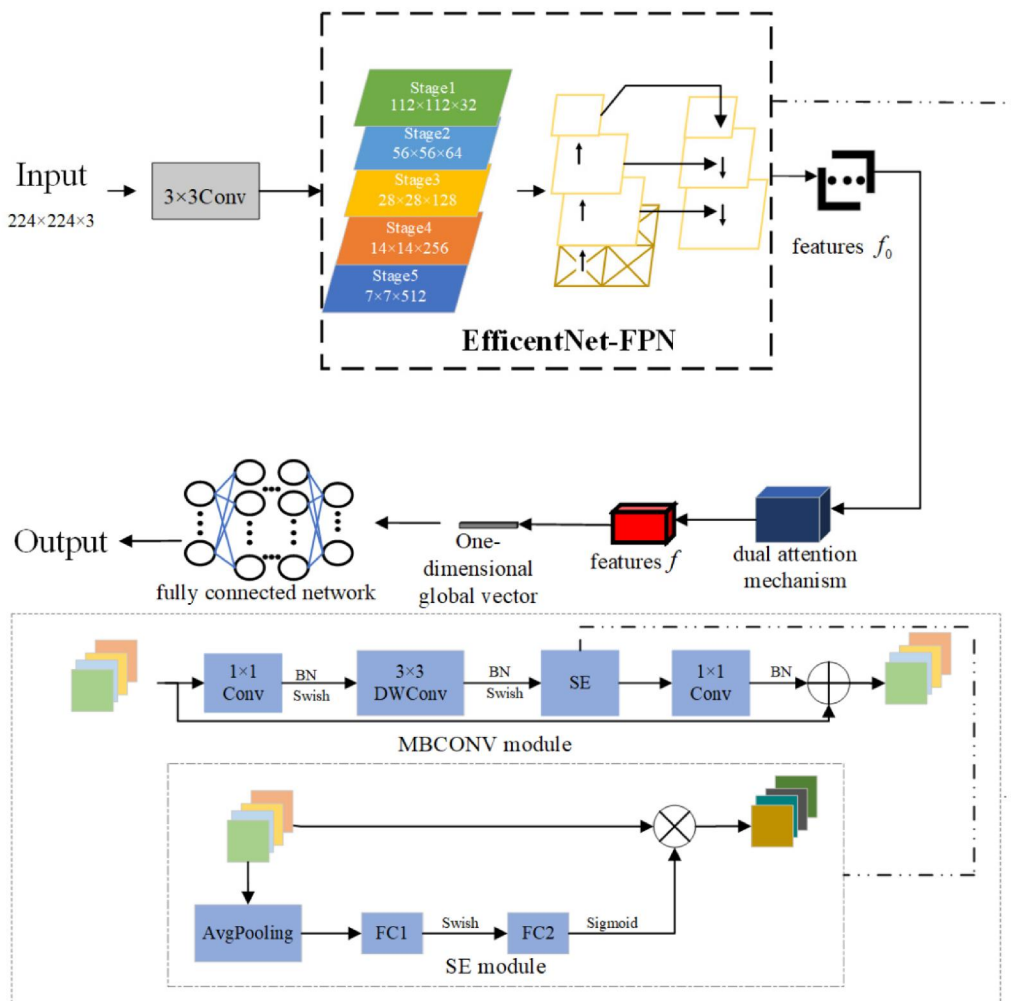


FIG. 3. Network diagram.

map, followed by dimensionality reduction using the first fully connected layer (FC1) and dimensionality expansion using the second fully connected layer (FC2). Subsequently, the features from both branches are fused. The second standard convolution then reduces the channel dimension, mapping the high-dimensional features back to the original feature space. Then, the input features and the processed features are summed to accomplish a residual skip connection. Additionally, residual connections are incorporated between each MBConv block to ensure network stability.

After extracting features from the EfficientNet backbone, the feature pyramid network (FPN) applies lateral convolution to feature maps from different stages to unify the channel dimensions. The high-level features are then progressively transmitted to low-level features through layer-by-layer upsampling and element-wise addition. The fused feature map is then refined by the convolutional layer. Finally, the feature map is spliced in the channel dimension to form a multi-scale feature representation of the same channel. The convolutional layer is fused to output the number of channels, which can be used as input features for subsequent modules.

2. Dual attention mechanism

Since the STL properties of MAM are extremely sensitive to the structural parameters, and the STL curves contain rich modal information and resonance features, the feature extraction capability of the network model directly affects the reliability of the prediction results. However, in traditional convolutional networks, it is easy to handle these very sensitive acoustic properties with insufficient attention to certain features, which affects the overall model's performance on the results. Therefore, a dual attention mechanism is introduced after the EfficientNet-FPN network to enhance the model's ability to capture critical acoustic features. As shown in Fig. 4, the dual attention mechanism transforms the feature tensor output from the EfficientNet-FPN network into new features through the channel attention mechanism (CAM) and spatial attention mechanism (SAM). The CAM and SAM operate in a tandem relationship, where the CAM is performed first, followed by the SAM.

The CAM processes the feature tensor output by the EfficientNet-FPN network and changes the size of the original features through two different operations, global average pooling and maximum pooling. It then compresses the number of channels

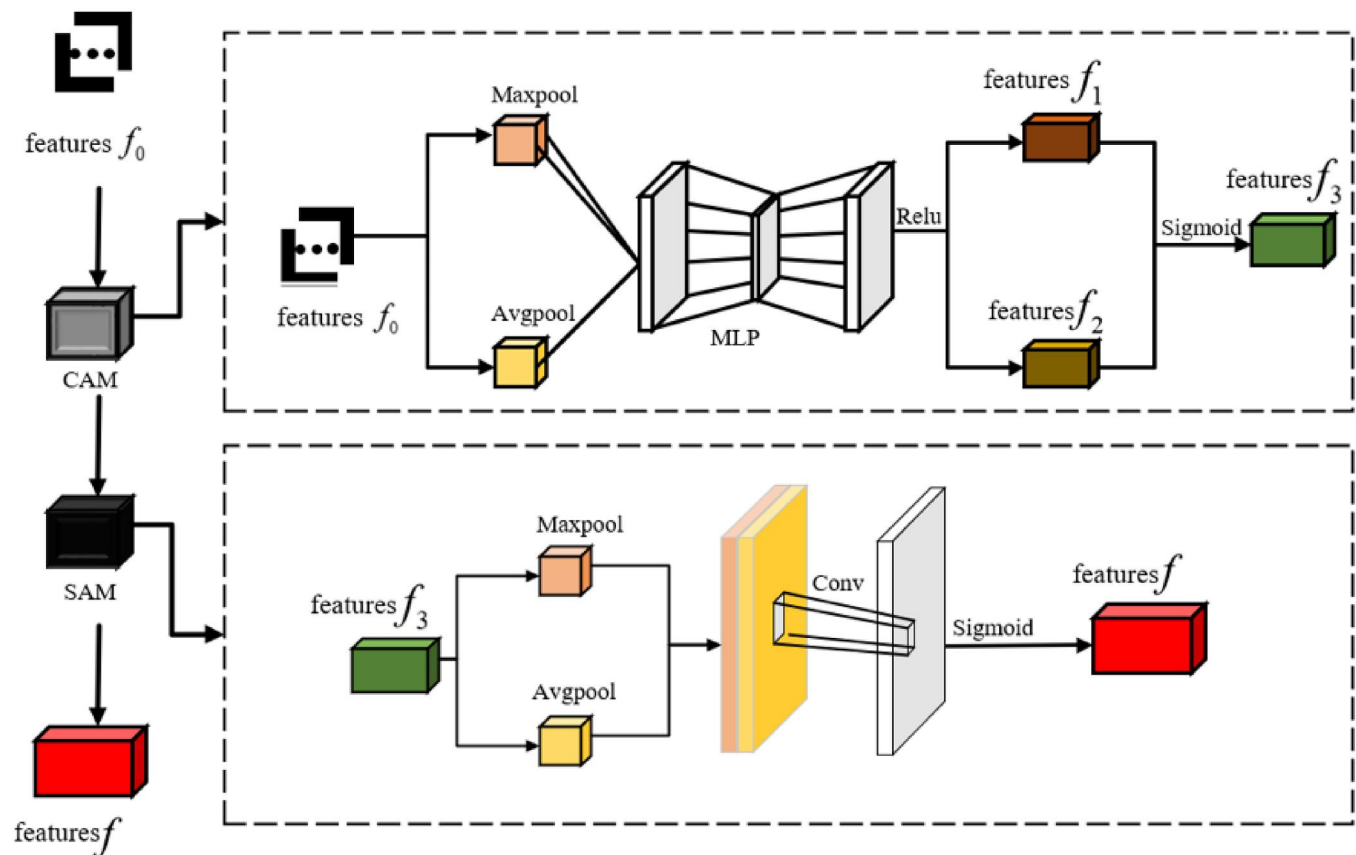


FIG. 4. Dual attention flowchart.

through the multilayer perceptron (MP) module and expands it to the original number of channels. Using the ReLU activation function, it obtains two different results and then sums the output of the two results element-by-element, generating the channel attention weights through the Sigmoid function. The SAM, on the other hand, performs average pooling and maximum pooling operations on the channel-weighted features to obtain two different spatial descriptions of the features. It then concatenates the inputs to the convolutional layer along the channel dimension, fuses the local and global spatial dependencies, and obtains the spatial attention weights after a Sigmoid function. Finally, it multiplies these weights with the original feature maps to restore the original size and outputs the new features. Finally, the feature maps enhanced by the dual attention mechanism are converted into one-dimensional vectors. The feature maps enhanced by the dual attention mechanism are finally transformed into one-dimensional vectors and fed into fully connected layers for multi-task prediction, including structural classification and parameter prediction. Finally, the network results are output.

B. The training module

1. Hyperparameter optimization

During network tuning, optimizing core hyperparameters like batch size and learning rate is crucial for achieving the best prediction performance. Manual adjustment based on experience is inefficient and costly. As illustrated in Fig. 5, we implement Bayesian optimization to globally search for the hyperparameter set that yields the lowest validation loss. The program supports two command-line execution modes. One initiates hyperparameter optimization, while the other bypasses it and proceeds directly to training. After optimization, the system prompts the user to confirm before entering the formal training phase.

2. Loss curves and ablation experiments

The loss function plays an important role in deep learning networks. For the multi-task learning problem of classification and regression tasks, using the same loss function will lead to unstable training and gradient mismatch problem. Therefore, cross-entropy loss is chosen to determine the probability distribution of the structure for the classification task, and Mean Squared Error (MSE) is used for the regression task to measure the deviation between the target value and the predicted value. The two different loss functions are weighted to obtain the weighted loss function, as shown in Eq. (21),

$$L = \lambda_1 L_{MSE} + \lambda_2 L_{Cross\ Entropyloss}. \quad (21)$$

The two task weights are automatically adjusted during training, and the network dynamically updates them by monitoring the loss variations of different tasks. This approach adaptively balances the two tasks during training, preventing any single task from dominating the network and degrading the final prediction performance. The optimal set of hyperparameters obtained based on Bayesian optimization is shown in Table IV, and the loss curve graph obtained after training is shown in Fig. 6. The graph shows

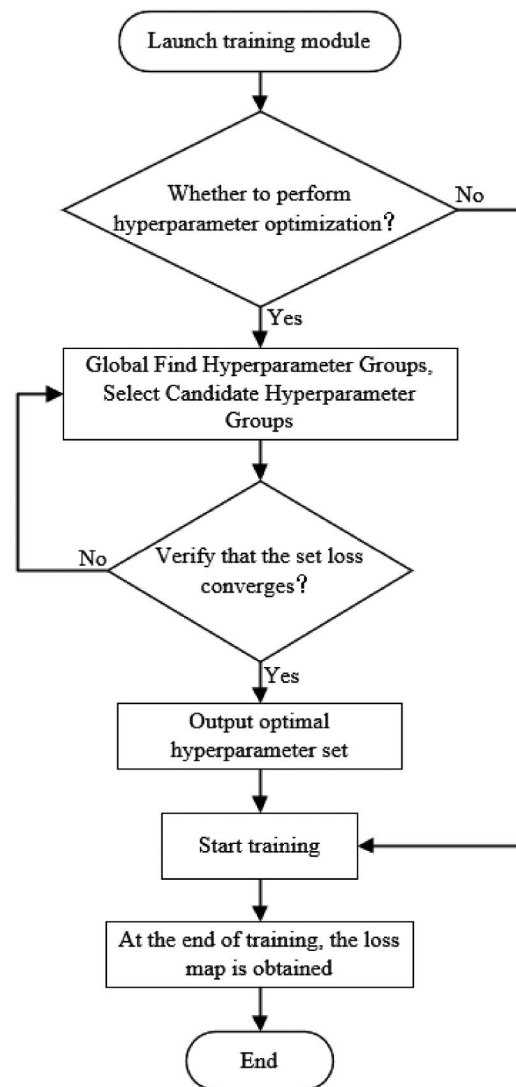


FIG. 5. Training module flowchart.

TABLE IV. Hyper parameter setting.

Hyperparameters	Value
Basic learning rate	0.003
Maximum learning rate	0.01
regularization factor	0.002
Batch size	16
Exponential moving average (EMA) initial attenuation rate	0.95
EMA final decay rate	0.999
Proportion of neurons randomly inactivated	0.45

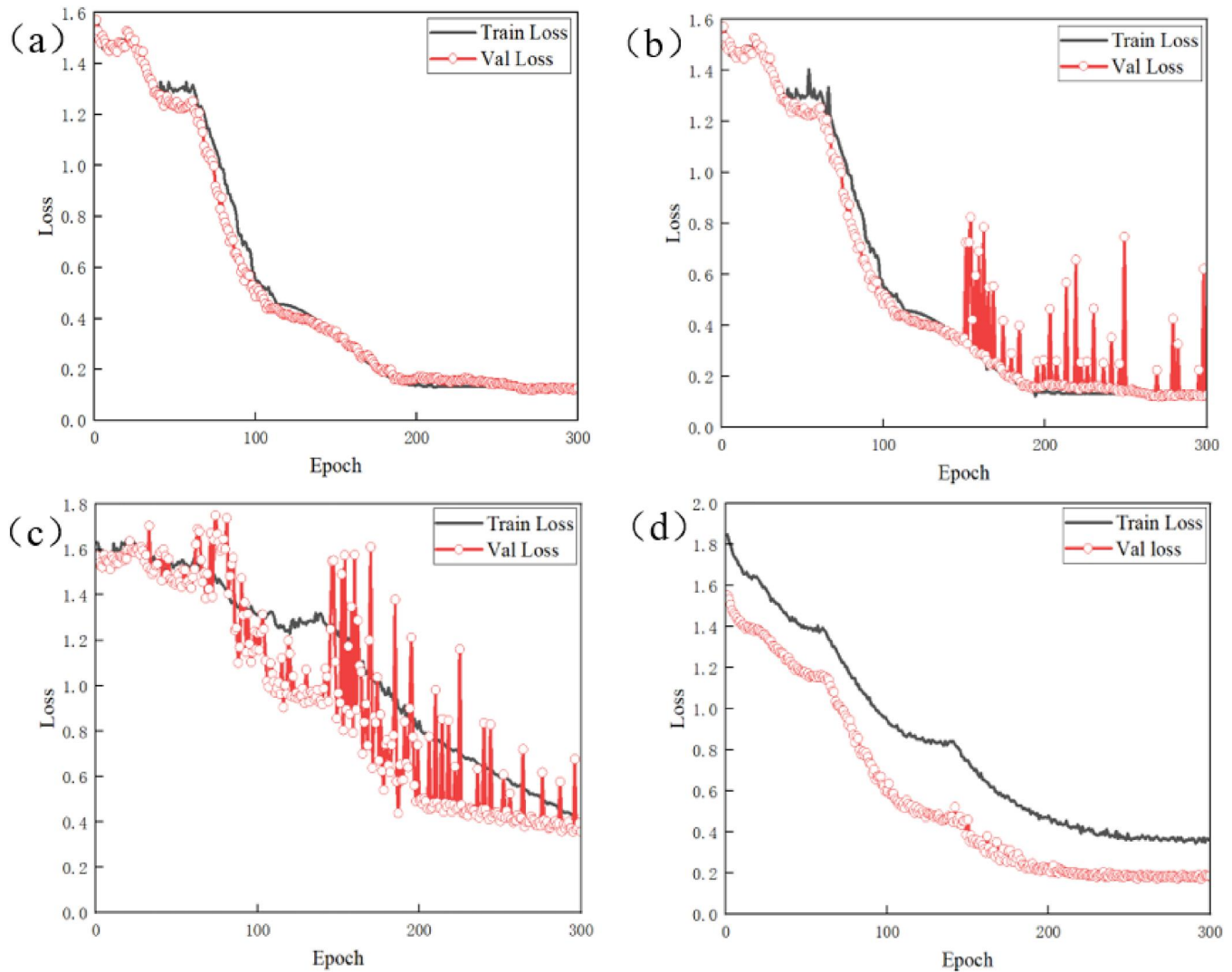


FIG. 6. Loss curve and ablation experiments. (a) FPN, dual attention, and automatic weighted loss function. (b) FPN. (c) Dual attention. (d) Fixed-weight loss.

that the fluctuations are more pronounced during the first 60 epochs because the two tasks exhibit different gradient distributions at the early stage of training, while the adaptive weight allocation has not yet stabilized. As a result, the synthetic gradient is prone to oscillation, leading to fluctuating curves and slow convergence. With the increase in the number of training rounds, after 200 rounds of training, the weight adaptation gradually stabilizes and the curve converges progressively.

To further evaluate the contributions of the FPN, dual attention module, and automatic weighted loss function, a combined ablation study was conducted under the same dataset, optimizer, and training settings. As shown in Fig. 6(a), the complete model with FPN, dual attention, and automatic weighted loss achieves stable convergence, with both the training loss and validation loss

decreasing smoothly to a low level. This indicates that the complete model has good training stability and generalization performance.

When only the FPN module is retained, as shown in Fig. 6(b), the loss curves still show an overall decreasing trend, suggesting that FPN is effective in extracting multi-scale features from STL curves. However, noticeable fluctuations and several spikes appear in the validation loss during the later training stage, indicating that FPN alone is insufficient to selectively emphasize key features and suppress unstable feature responses. When only the dual attention module is used, as shown in Fig. 6(c), the convergence becomes slower, and both the training loss and validation loss exhibit stronger oscillations. This indicates that, although dual attention can enhance important channel and spatial features, the

absence of multi-scale feature fusion limits the model's ability to represent complex STL curve characteristics. In addition, Fig. 6(d) shows the fixed-weight loss baseline. Although the fixed-weight model can also converge, its training loss remains generally higher and decreases more slowly in the middle and later stages. Its final validation loss is also higher than that of the automatic weighted model in Fig. 6(a). Quantitatively, the optimal validation loss of the automatic weighted model is 0.1368, which is lower than the 0.1740 obtained by the fixed-weight model, corresponding to a reduction of approximately 21.4%. These results demonstrate that the combination of FPN and dual attention improves feature extraction and training stability, while the automatic weighted loss function further balances the structural classification and parameter regression tasks, thereby improving the overall convergence performance and prediction accuracy of the inverse design model.

C. The prediction module and results

The core of the prediction module consists of two parts, the STL curve generator and the parameter inversion predictor, which are connected in series. The prediction module is used to validate the neural network model after the training is completed, which is the key part to test the generalization ability of the network model. The prediction module first generates several STL curves that satisfy the target acoustic requirements, and the network model then predicts the corresponding structural parameters from these generated curves.

To verify the prediction accuracy of the model, the input to the prediction module is a specified STL diagram, and the prediction of structure type and material parameters is performed on the STL diagram. The original and predicted parameter sets are shown in Table V. The percentage of structure type and the percentage of parameter error are shown in Fig. 7. From Fig. 7(a), the percentage of the three groups predicted to be structure 1 is above 50%. From Figs. 7(b) and 7(c), the predicted membrane density and rubber layer density have an error between 3.6%–4.5% and 3%–8.93% from the original parameters. From Figs. 7(d) and 7(e), it can be seen that the errors of the predicted mass-block radius and the membrane thickness with the original parameters are in the range of 2%–6% and 6.5%–10.5%, which indicates that the designed network has a high prediction accuracy. Figure 8 compares the predicted and original STL curves, and Table VI lists the peak STL values and corresponding frequencies of the first and second peaks. As shown in Table VI, the peak-frequency errors of the three predicted samples range from 5 to 20 Hz, while the STL errors range from 1.13 to 10.92 dB. The maximum frequency error is 20 Hz, occurring in the first predicted sample, and the

maximum STL error is 10.92 dB, occurring at the first peak of the second predicted sample. Overall, the predicted curves show good agreement with the original STL curves in terms of the main peak positions and peak amplitudes. Although some deviations remain in the peak STL values of the predicted curves, Fig. 8 indicates that the three predicted STL curves remain in good agreement with the original curve in terms of both overall trend and peak positions. These results indicate that the proposed neural network model can effectively capture the key acoustic characteristics of the STL response, thereby demonstrating the reliability of the inverse prediction framework.

To further evaluate the performance of the reverse design model, the coefficient of determination R^2 is incorporated to assess model accuracy. The R^2 coefficient can be expressed as

$$R^2 = 1 - \frac{\sum_{i=1}^n (STL_i^{ture} - STL_i^{pred})^2}{\sum_{i=1}^n (STL_i^{ture} - \overline{STL}^{ture})^2}, \quad (22)$$

where STL_i^{ture} denotes the sound insulation value at the i frequency point for the actual measurement, STL_i^{pred} denotes the sound insulation value at the i frequency point for the predicted measurement, n represents the number of frequency sampling points, and $\overline{STL}^{ture}$ denotes the actual STL average value. Figure 9 illustrates the comparison between parameter values predicted by the reverse design model and actual parameters. Within this figure, a small number of data points exhibit a certain deviation relative to the 45° reference line, a phenomenon primarily occurring at the peaks or troughs of the sound insulation curve. This arises because minute variations in material parameters cause shifts in resonance frequency, resulting in deviations of several hertz between the predicted STL and the true STL at these peak and trough values. Nevertheless, the overall trend reveals that the majority of data points cluster closely around the dashed line, accompanied by a high coefficient of determination. This indicates a high degree of agreement between the actual and predicted values.

D. Physical mechanism and reliability analysis

1. Displacement field and physical mechanism analysis

To clarify the physical mechanism of the predicted structures, displacement-deformation analyses were conducted at the representative STL valley and peak frequencies, as shown in Fig. 10. For the first predicted structure, Figs. 10(a)–10(c) show the displacement fields at 205, 375, and 440 Hz, respectively. At 205 Hz, the

TABLE V. RMMAM specifies and predicts parameter values.

	Structure	R_1 and R_2 (mm)	ρ_1 (kg/m ³)	ρ_2 (kg/m ³)	ρ_3 (kg/m ³)	d (mm)
Original data	1	2.0	8960	1600	1600	0.2
Predictive data 1	1	2.12	8960	1457	1541	0.187
Predictive data 2	1	1.93	8960	1521	1662	0.231
Predictive data 3	1	2.04	7870	1648	1528	0.183

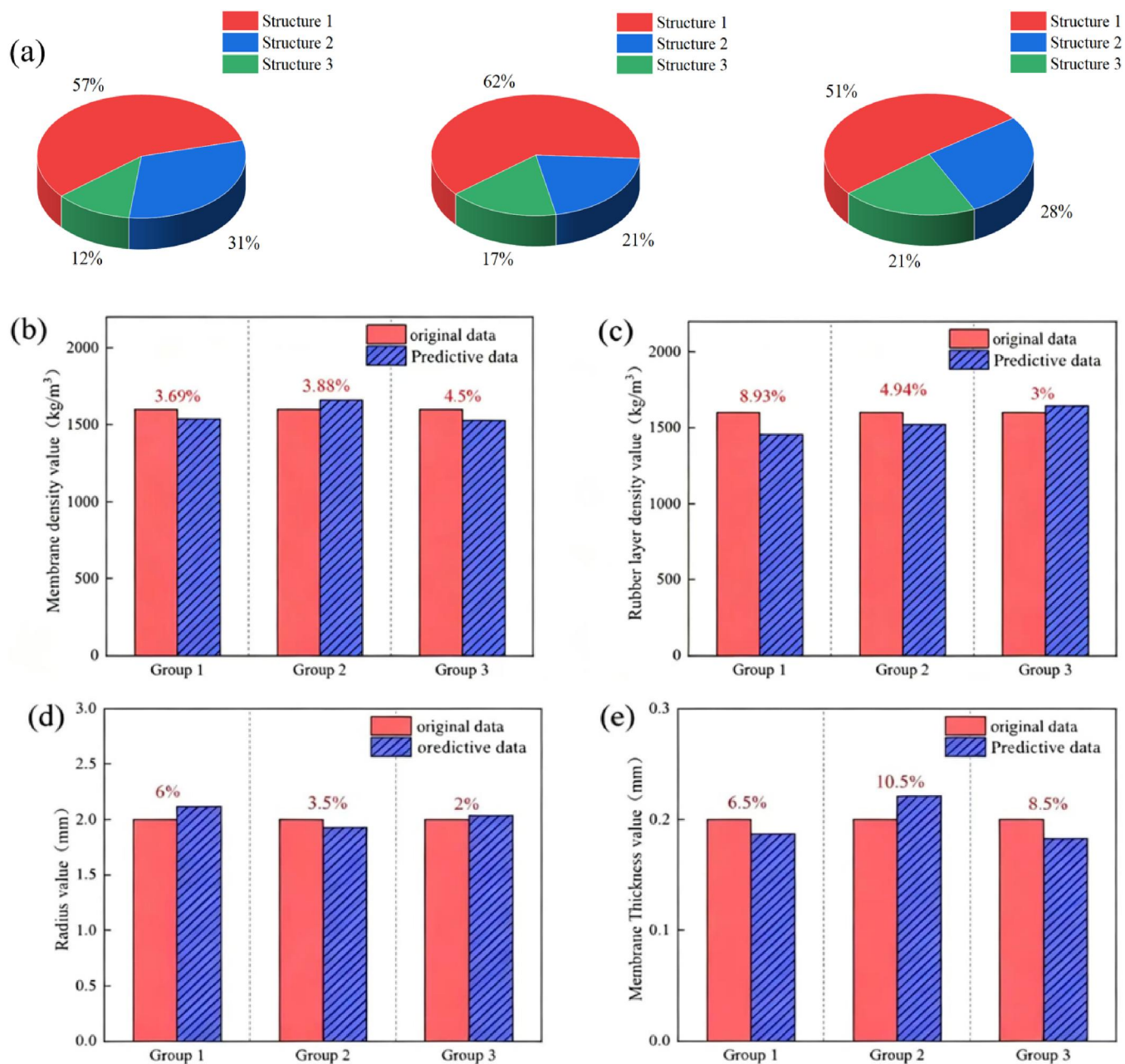


FIG. 7. Prediction of structure type and material parameters. (a) Percentage of each structure. (b) Membrane density error. (c) Rubber layer density error. (d) Radius error. (e) Membrane thickness error.

membrane and mass blocks mainly move in phase, resulting in weak vibration suppression and the STL valley. At 375 Hz, the central mass block exhibits a strong localized response and moves out of phase with the membrane, increasing the equivalent acoustic impedance and forming the first STL peak. At 440 Hz, the dominant vibration shifts to the diagonal mass blocks,

corresponding to a higher-order localized coupling mode and the second STL peak. For the second predicted structure, Figs. 10(d)–10(f) show similar valley-peak evolution. The in-phase motion at 170 Hz leads to easier sound transmission, while the out-of-phase motion at 300 Hz enhances vibration constraint and produces the first STL peak. At 370 Hz, the displacement is mainly

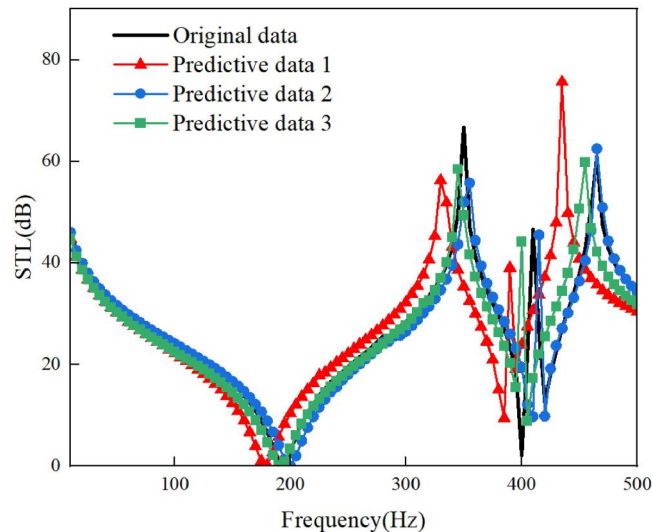


FIG. 8. STL comparison of original and predicted parameters.

concentrated around the central mass block, indicating a localized vibration mode that suppresses membrane vibration transmission and forms the second STL peak. The third predicted structure in Figs. 10(g)–10(i) exhibits a displacement pattern similar to that of the first structure. Overall, the STL valleys are mainly associated with in-phase motion between the membrane and mass blocks, whereas the STL peaks originate from out-of-phase motion and localized coupling vibration. This confirms that the predicted STL characteristics are physically related to membrane-mass coupling, indicating that the model captures key structural features affecting sound insulation performance.

2. Sensitivity analysis of structural and material parameters

As can be seen from Fig. 11, the original structural parameters are $R_1 = R_2 = 2.0$ mm, $d = 0.2$ mm, $\rho_2 = 1600$ kg/m³, and $\rho_3 = 1600$ kg/m³. Figure 11 shows the sensitivity analysis of STL

TABLE VI. Comparison of peak sound insulation values and peak frequencies between the original and predicted parameters.

	Peak position	Frequency (Hz)	Error	Sound insulation (dB)	Error
Original data	First peak	350		66.67	
	Second peak	410		46.65	
Predictive data 1	First peak	330	20	56.65	10.02
	Second peak	390	20	40.05	6.6
Predictive data 2	First peak	355	5	55.75	10.92
	Second peak	415	5	45.52	1.13
Predictive data 3	First peak	345	5	58.48	8.19
	Second peak	400	10	44.24	2.41

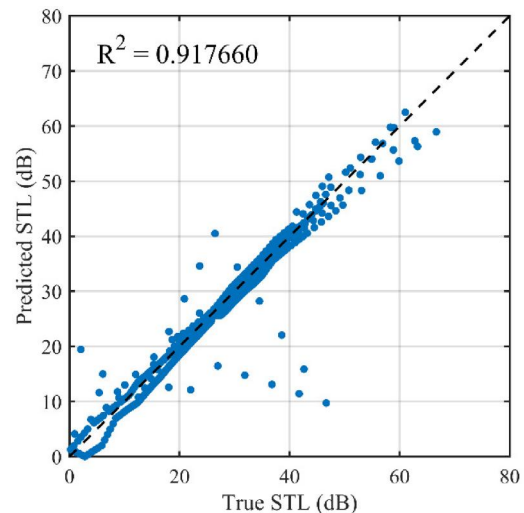


FIG. 9. Comparison of true STL and predicted STL.

curves under variations in structural and material parameters. The radius and membrane thickness were varied by 10%, while the two density parameters were varied by 12.5%. As shown in Fig. 11(a), changing the composite mass-block radius has little effect on the low-frequency valley near 200 Hz, and the overall STL trend remains similar. Only slight shifts appear in the peak region, indicating that the radius mainly affects the local resonance intensity rather than the basic sound insulation mechanism. In Fig. 11(b), the membrane thickness has a stronger influence. With the increasing thickness, the STL peaks shift to higher frequencies, especially the second peak, because the increased thickness raises the equivalent stiffness of the membrane-mass system. Figures 11(c) and 11(d) show the effects of rubber mass-block density and membrane density, respectively. Changes in density have limited influence on the low-frequency valley, but affect the amplitudes and positions of the high-frequency peaks by changing the equivalent mass and localized resonance strength. Within the investigated parameter variation range of 10%–12.5%, the STL curves still maintain the main low-frequency valley and double-peak characteristics, without obvious failure or severe deviation. This indicates that the predicted structures have a certain robustness to structural and material parameter errors. Consequently, although the maximum error in structural parameters reported in this study reached 10.5%, the predicted structure retained the main STL peak-to-trough characteristics and basic sound insulation performance within the range of parameter variations studied, demonstrating that this level of error is acceptable.

IV. INVERSE DESIGN OF RMMAM WITH CONSTRAINED ACOUSTIC PROPERTIES

A. Network design and analysis of results

To demonstrate the effectiveness of this inverse design methodology in engineering applications, a case study was conducted.

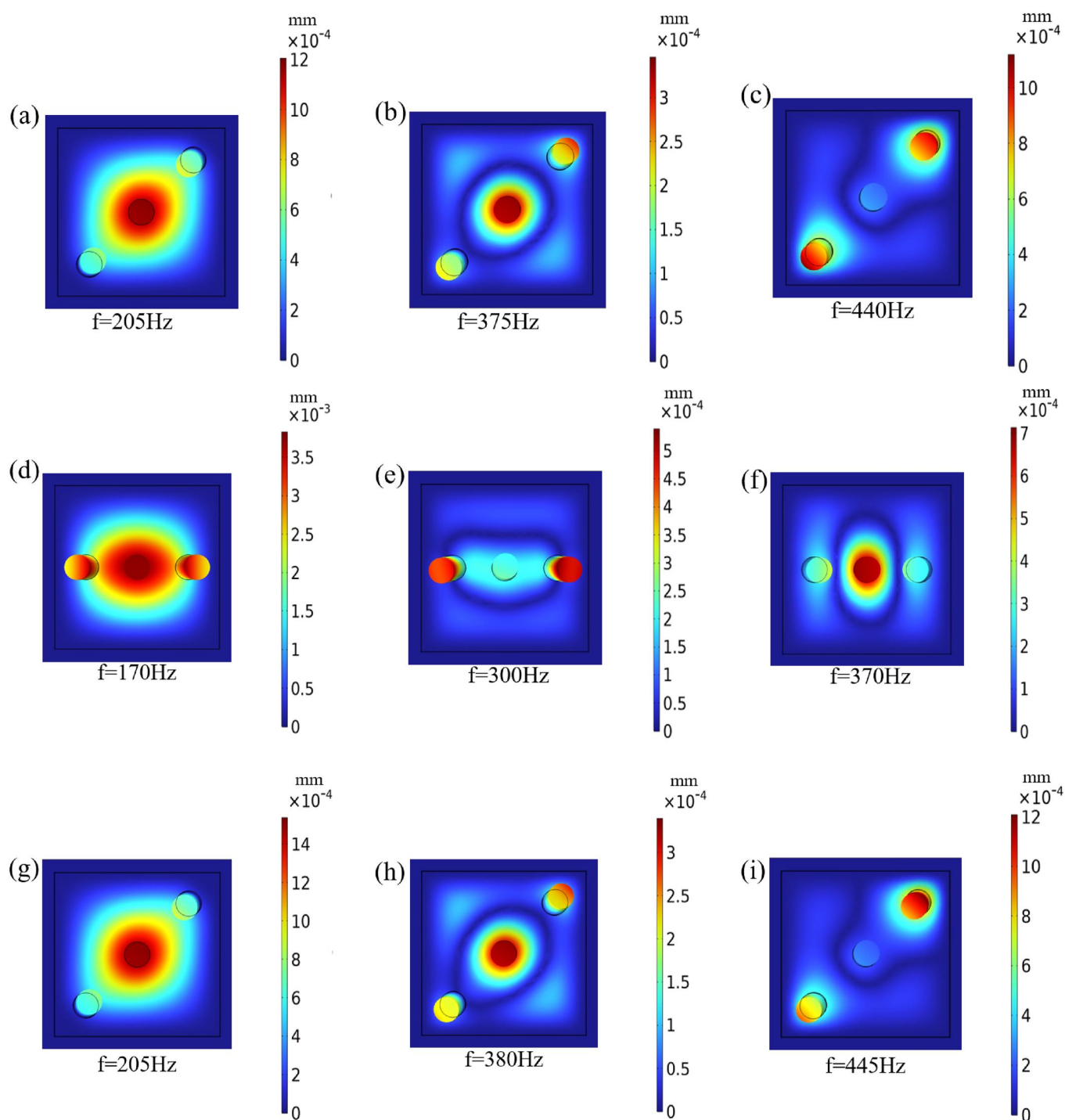


FIG. 10. Displacement-deformation diagram. (a) The first trough in predictive data 1. (b) The first peak in predictive data 1. (c) The second peak in predictive data 1. (d) The first trough in predictive data 2. (e) The first peak in predictive data 2. (f) The second peak in predictive data 2. (g) The first trough in predictive data 3. (h) The first peak in predictive data 3. (i) The second peak in predictive data 3.

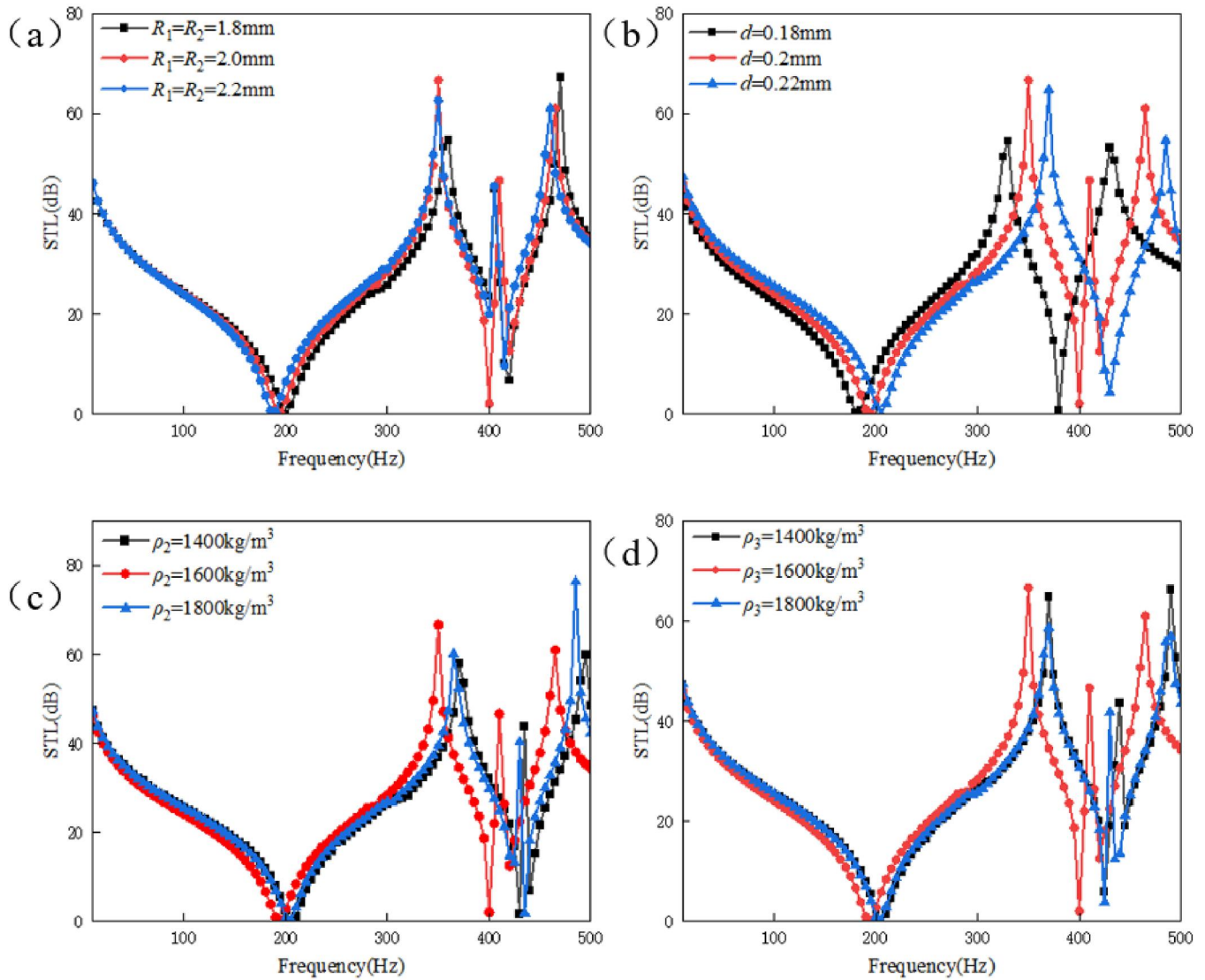


FIG. 11. Sensitivity analysis of STL curves with respect to variations in structural and material parameters. (a) Radius of a composite mass block. (b) Membrane thickness. (c) Density of rubber blocks. (d) Density of membrane.

The study aimed to meet an acoustic performance constraint, achieving a peak STL of no less than 50 dB within the 300–400 Hz frequency band. The prediction process is illustrated in Fig. 12. The prediction module first employs the STL curve generator to randomly generate candidate STL curves that satisfy the prescribed acoustic performance constraints. Subsequently, all the candidate curves are converted to an image format consistent with the training stage and fed into the trained EFDA-CNN for structure type prediction and parameter prediction. The scoring mechanism [as shown in Eq. (23)] is introduced to select the three groups of parameters with the highest scores. The three groups of parameters are shown in Table VII. The obtained STL curves are shown

in Fig. 13, and all three curves satisfy the performance limit of 50 dB STL in the range of 300–400 Hz,

$$Q = 0.5 \left(1 - \frac{|\Delta x|}{\Delta x_{\max}} \right) + 0.5 \frac{\Delta d}{\Delta d_{\max}}, \quad (23)$$

where Δx denotes the difference between the actual frequency and the target frequency and $\Delta x_{\max}$ denotes the maximum difference between the actual frequency and the target frequency. Δd denotes the difference between the actual sound insulation and the target

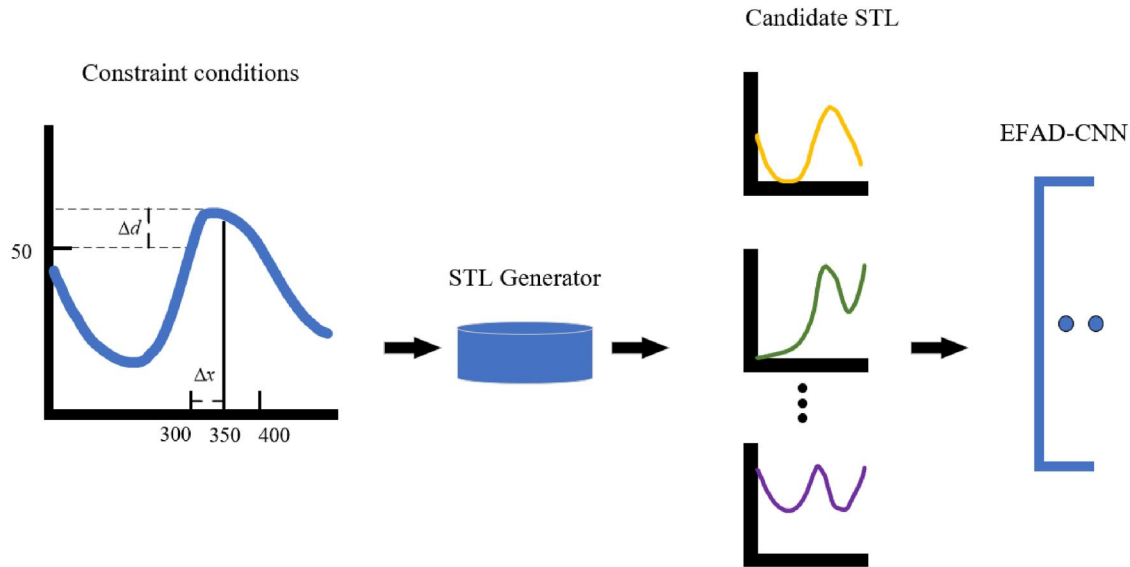


FIG. 12. Forecasting process.

STL, and $\Delta d_{\max}$ denotes the maximum difference between the actual sound insulation and the target STL.

B. Experimental results and discussion

An impedance tube test was performed in a semi-anechoic chamber to validate the prediction results. A schematic of the setup is shown in Fig. 14, including the impedance tube, the data acquisition device, microphones, the computer, and the test sample. The impedance tube system consists of a loudspeaker, an incident chamber, and a transmission chamber. A thick tube with a diameter of 100 mm was used for the experiment. The microphone is the B&K 4187 type with a sensitivity deviation range of ± 0.3 dB. The microphone needs to be calibrated before the test, and each microphone is spaced 50 mm apart. The experimental samples are made according to the parameters in Table VII, as shown in Fig. 15. The frame material is acrylonitrile butadiene styrene (ABS) resin, the membrane material is high

density polyethylene (HDPE) (polyethylene) membrane, the lower mass-block material is rubber, and the upper mass-block material is metal.

The results of the impedance tube experiment and finite element simulation are compared and verified, and Table VIII shows that the first peak-frequency errors of samples A, B, and C are 45, 60, and 75 Hz, the second peak-frequency errors are 40, 20,

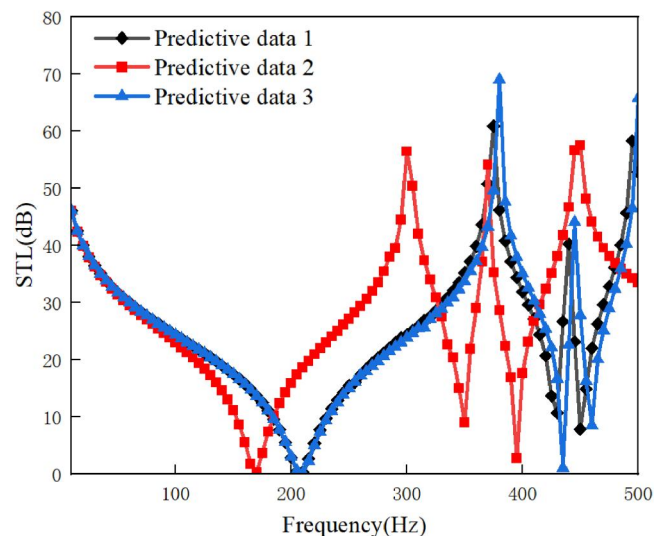


FIG. 13. STL for prediction parameters.

TABLE VII. Predicted parameter results.

	Structure	R_1 and R_2 (mm)	ρ_1 (kg/m ³)	ρ_2 (kg/m ³)	ρ_3 (kg/m ³)	d (mm)
Predictive data 1	1	1.81	7870	1419	1237	0.221
Predictive data 2	2	2.06	8960	1192	1580	0.293
Predictive data 3	1	1.92	7140	1896	1598	0.214

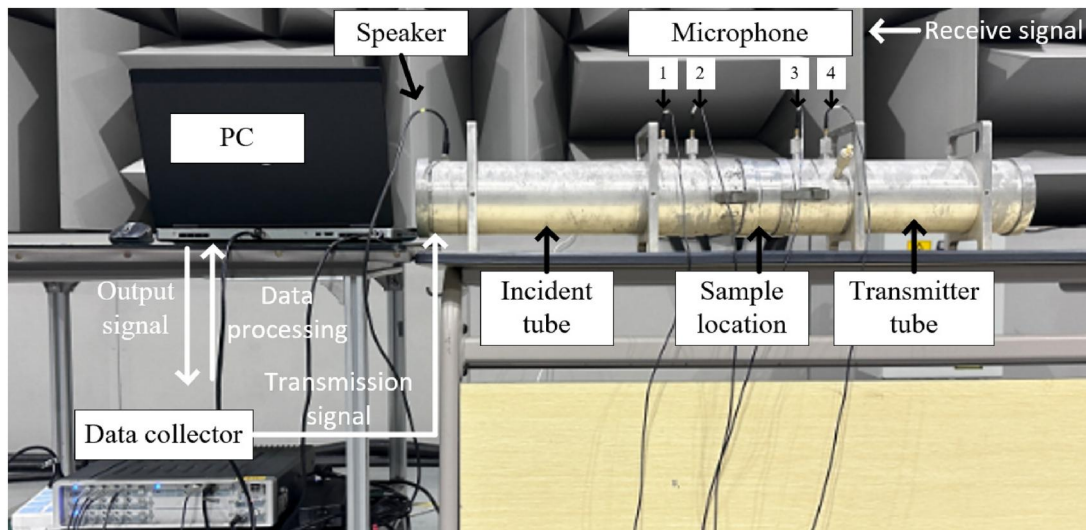


FIG. 14. Experimental setup.

and 45 Hz, the first peak STL errors are 6.94, 5.12, and 1.57 dB, and the second peak STL errors are 0.75, 13.64, and 11.2 dB. As can be seen from Fig. 16, the impedance tube experiments and the finite element simulation maintain a high degree of consistency in the curve trends, and the STL curves satisfy the acoustic performance constraint of STL of not less than 50 dB within 300–400 Hz.

However, discrepancies were observed between the experimentally measured STL and the simulated STL. The maximum STL error is 13.64 dB, which mainly occurs near the second sound insulation peak, whereas the experimental and simulated results agree well near the first peak. These discrepancies can be attributed to several factors.

First, the actual sample boundary condition is different from the ideal finite element model. In the simulation, the sample is assumed to be perfectly sealed in the impedance tube. In the experiment, however, the sample diameter is slightly smaller than

that of the tube fixture, leaving small gaps near the edge. Although petroleum jelly was used for sealing, slight acoustic leakage still exists and reduces the measured STL peak. Second, sample fabrication and assembly also introduce errors. Because the composite mass blocks were manually attached, their actual positions may slightly differ from those in the finite element model. This changes the local coupling between the mass blocks and the membrane and, thus, affects the resonance peak frequency and amplitude. In addition, the relatively large sample frame introduces extra damping and changes the local vibration response. Third, the experimental constraints are not fully identical to those in the simulation. In the finite element model, fixed constraints are applied in all three directions, whereas an equivalent fixed constraint in the Z direction is difficult to achieve in the impedance tube test. This may induce additional out-of-plane vibration and boundary disturbance, especially in the high-frequency peak region.

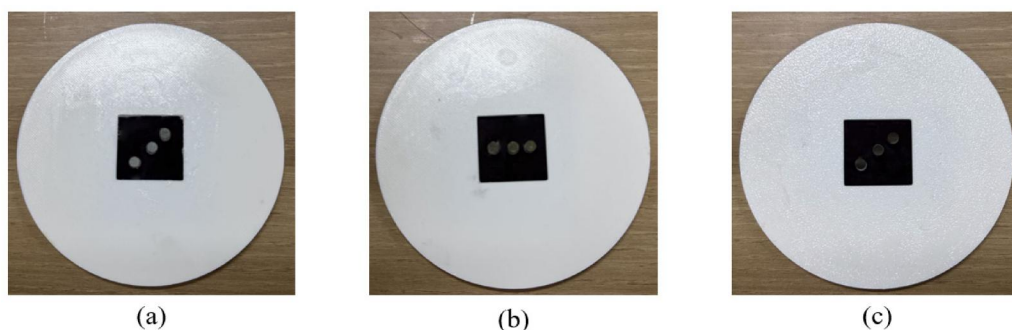


FIG. 15. Experimental sample of RMMAM. (a) Sample A (metal layer is iron). (b) Sample B (the metal layer is copper). (c) Sample C (the metal layer is zinc).

TABLE VIII. Comparison of data from sample A, sample B, and sample C.

Sample	Peak position	Peak sound insulation frequency (Hz)			Peak sound insulation (dB)		
		FEM	Experiment	Error	FEM	Experiment	Error
A	First peak	375	330	−45	60.76	53.82	−6.94
	Second peak	440	400	−40	40.26	39.51	−0.75
B	First peak	300	240	−60	56.48	51.36	−5.12
	Second peak	370	350	−20	54.18	40.54	−13.64
C	First peak	380	305	−75	69.05	67.48	−1.57
	Second peak	445	400	−45	44.09	32.89	−11.2

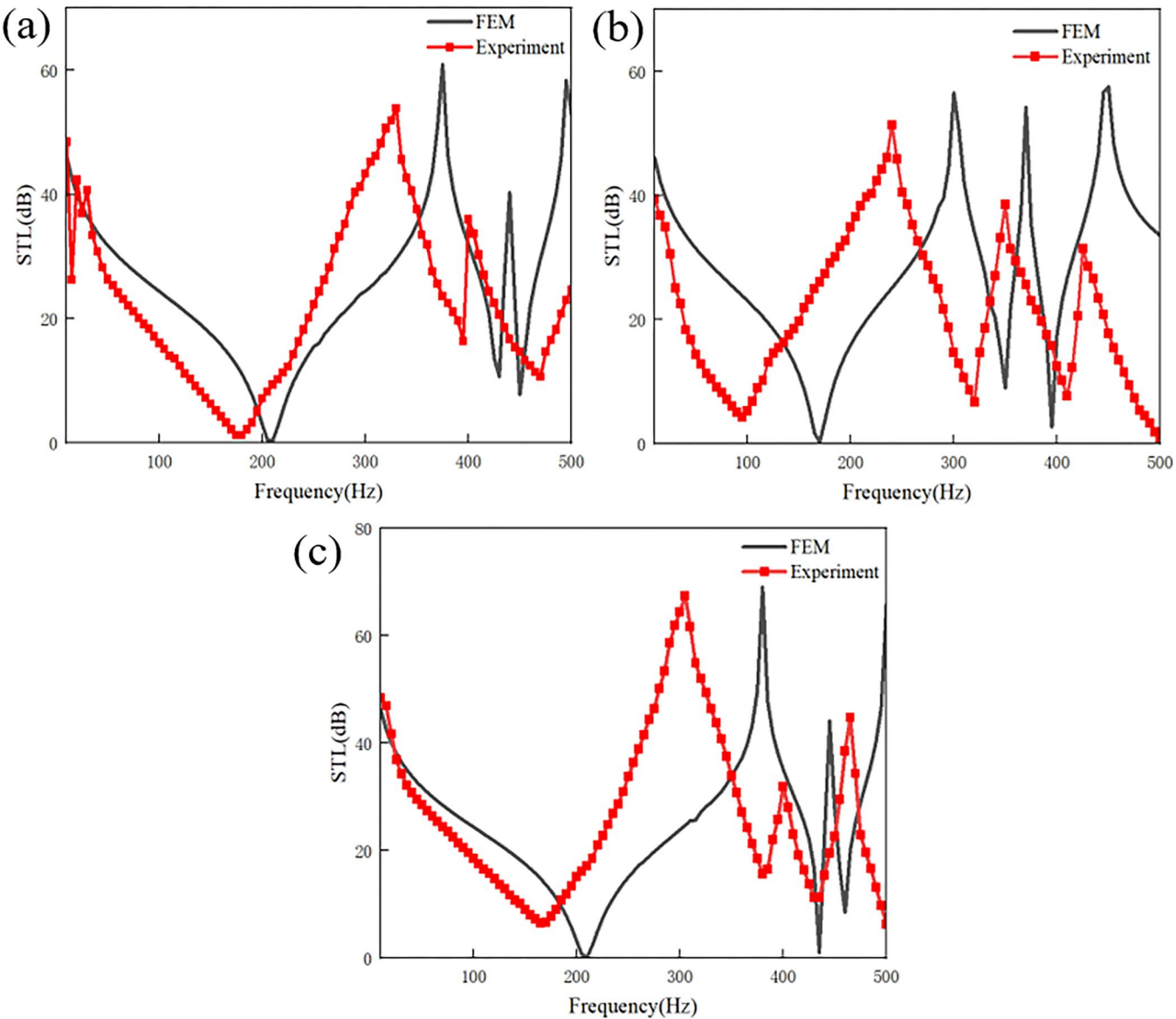


FIG. 16. Comparison of simulated and experimental STL. (a) Sample A. (b) Sample B. (c) Sample C.

V. CONCLUSION

In this paper, an EFDA-CNN network model is proposed with the aim of solving the problem related to the inverse design of structural types and material parameters of MAM based on acoustic property constraints. The network model consists of EfficientNet-FPN as the backbone network combined with the dual attention mechanism. To verify the prediction accuracy of the EFDA-CNN network model, the specified STL acoustic isolation graphs were analyzed as inputs, and the results showed that the relative errors in the prediction of the membrane density and the density of the rubber layer were controlled within the ranges of 3.6%–4.5% and 3.0%–8.93%, respectively. The prediction errors of geometric parameters such as mass-block radius and membrane thickness were kept within the ranges of 2.0%–6.0% and 6.5%–10.5%, respectively. Meanwhile, to illustrate the practical engineering significance of this reverse design system, the case study of achieving an STL of 50 dB in the frequency band of 300–400 Hz is presented. Finally, through the verification comparison of impedance tube experiments and finite element simulations, the results show that the three sets of parameters predicted by the surface prediction design are able to achieve the set constraints of sound insulation performance, which proves the feasibility and accuracy of the RMMAM reverse design network.

At the same time, the automated co-simulation workflow established between COMSOL and MATLAB via LiveLink enables the rapid generation of finite element datasets, providing an efficient approach for dataset construction in data-driven studies of MAMs and other multi-parameter coupled structures. In addition, the proposed inverse design method can directly predict the structural type and parameter combination from target STL curves, making it well suited for the rapid design of structures with specific sound insulation requirements and giving it potential application value in construction, aerospace, and transportation. However, the current model is still limited to three predefined structural types. When the target STL curve falls within their response range, the model can provide effective predictions; when it lies outside this range, the model can only search for the closest structure from the existing templates, which may result in incomplete matching between the acoustic characteristics and the target response. This limitation could be improved by expanding the design space and constructing a larger and more diverse dataset. In addition, non-unique solutions may still occur when different structures produce similar acoustic responses. Therefore, future work will focus on enlarging the design space and introducing physics-informed neural networks to improve the reliability and generalization capability of the model.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

ShengLin Xu: Conceptualization (equal); Data curation (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Shuang Huang:** Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – original draft (equal). **Hui Guo:** Funding acquisition (equal); Resources (equal). **Yansong Wang:** Funding acquisition (equal); Resources (equal); Supervision (equal). **pei sun:** Methodology (equal); Software (equal); Supervision (equal). **Ningning Liu:** Methodology (equal); Software (equal); Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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